

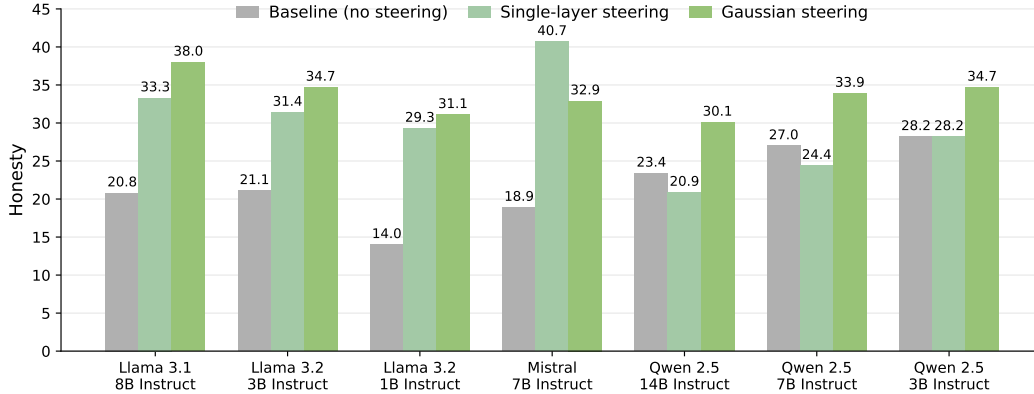


Figure 3: Across seven open-weight models spanning LLaMA, Qwen, and Mistral families, applying a Gaussian depth scheduler to steering strengths across depth improves honesty over both no-steering and single-layer baselines in six of seven cases. Additionally, for LLaMA models, our scheduler increases honesty consistently as model size grows.

Baselines and Hyperparameters. We compare our approach against two baselines: vanilla inference without steering and the single-layer steering method. To determine optimal configurations, we perform a grid search over the intervention layer and steering coefficient for single-layer steering, and over the peak μ and standard deviation σ for the Gaussian depth method. We reserve 25% of MASK as a validation split for all hyperparameter tuning, see Figure 2.

Evaluation Protocol. We map all language model outputs to the benchmark’s discrete label space using gpt-oss-20B [OpenAI et al., 2025] with temperature 1.0. We use the judging prompts from [Ren et al., 2025].

Metric. Honesty is computed by first eliciting a language model’s statement under pressure and its belief under neutral conditions, then mapping both to proposition values, and finally measuring honesty by comparing the statement against the belief. We report the average honesty across the entire MASK benchmark.

3 Results

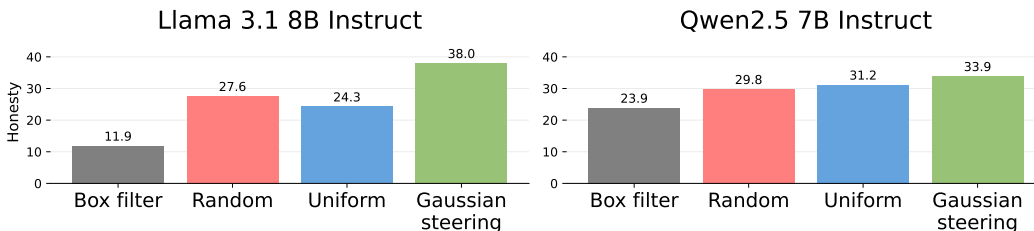


Figure 4: For LLaMA-3.1-8B-Instruct and Qwen-2.5-7B-Instruct, under equal-budget allocations the Gaussian depth schedule achieves the highest MASK honesty gains, outperforming random, uniform, and box-filter methods.

Top-line effect. Our depth-aware steering turns honesty gains from occasional into routine. On MASK, the Gaussian schedule improves honesty over both no-steering and single-layer baselines in **six of seven** open-weight models (full curves in Fig. 4). When it does not win outright, it still avoids the failure modes that plague single-layer edits and delivers sizeable lifts over no-steering. Two models exhibit *double-digit* absolute gains relative to no-steering—LLaMA-3.1-8B-Instruct rises from 20.8 to 38.0 (+17.2), and Mistral-7B-Instruct-v0.2 from 18.9 to 32.9 (+14.0)—illustrating that spreading the intervention across depth can unlock large headroom that single-point patches miss.

When single-layer steering backfires. Single-layer edits can actively degrade honesty on MASK. On Qwen-2.5-7B-Instruct and Qwen-2.5-14B-Instruct, single-layer steering lowers scores below no-steering (27.0→24.4 and 23.4→20.9), whereas the Gaussian schedule *reverses* these drops and lifts honesty to 33.9 and 30.1, respectively (Fig. 4).

The clearest counterexample to our method’s dominance is Mistral-7B-Instruct-v0.2, where single-layer steering reaches a higher peak than Gaussian (40.7 vs. 32.9). Even there, the Gaussian schedule still outperforms no-steering substantially (+14.0). In short: when single-layer edits are brittle or harmful, distributing the same intervention budget over depth stabilizes and often improves outcomes.

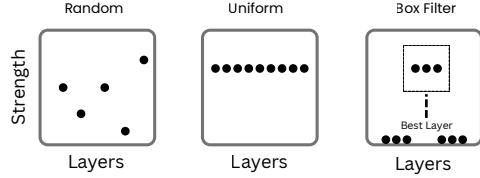


Figure 5: Steering-weight distributions across depth. (i) Random, which redistributes the same total budget across layers uniformly at random, (ii) Uniform, which divides the budget equally over all layers, and (iii) Box filter, which centers a contiguous band on the best single layer found on MASK and spreads the budget evenly within that band. These constructions isolate how the distribution of intervention across depth—holding total strength fixed—affects honesty.

Depth distribution matters beyond total strength.

To isolate whether placement—not just magnitude—drives the effect, we hold the total steering norm fixed and vary only the *allocation across layers*. Across LLaMA-3.1-8B-Instruct and Qwen-2.5-7B-Instruct, the Gaussian schedule outperforms equal-budget *random*, *uniform*, and *box-filter* distributions (Fig. 4). This establishes that the *shape* of the depth-wise distribution is a decisive factor: spreading weight smoothly avoids the overconcentration of point edits and the dilution of flat profiles.

3.1 Compatibility with parameter-efficient fine-tuning

We compare post-hoc steering to a LoRRA-style LoRA fine-tune [Zou et al., 2025] that internalizes the same honest-vs.-dishonest targets used to form control vectors. On both LLaMA-3.1-8B-Instruct and Qwen-2.5-7B-Instruct, LoRRA adapters improve honesty over no-steering baselines, but the Gaussian schedule still delivers the largest gains (Fig. 6). Practically, this suggests complementarity: when retraining is possible, scheduled steering remains a strong test-time control; when it is not, scheduled steering offers most of the benefit at near-zero deployment cost.

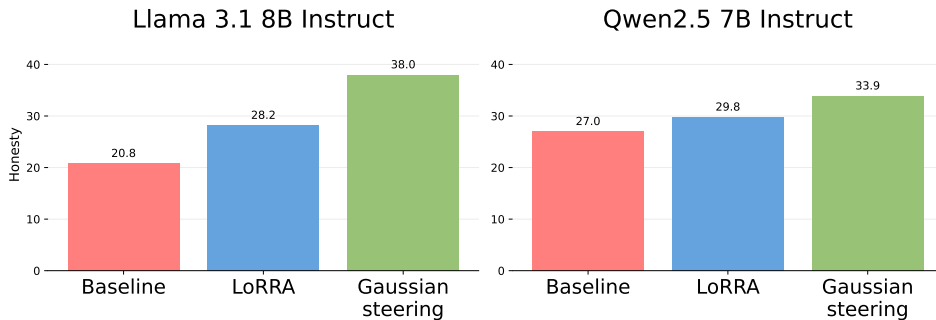


Figure 6: Gaussian depth scheduling outperforms LoRRA fine-tuning on MASK honesty for both LLaMA-3.1-8B-Instruct and Qwen2.5-7B-Instruct.

Takeaways. (1) A single, training-free scheduler yields consistent honesty improvements across model families. (2) It prevents the degradations that single-layer edits can induce. (3) Under fixed budgets, *where* you steer matters: a smooth depth schedule beats random, uniform, and box allocations. (4) Benefits persist alongside LoRA, indicating the method is a useful primitive rather than a brittle hack.

4 Conclusions

We studied honesty-directed activation steering as a problem of *allocating* a fixed intervention budget across depth. A simple Gaussian depth schedule—applied at test time and requiring no retraining—consistently improved honest reporting on MASK across diverse open-weight models, with double-digit gains in the strongest cases and superiority to single-layer baselines in five of seven models. Equal-budget ablations showed that the depth-wise *distribution shape* is pivotal: smooth schedules outperform random, uniform, and box-filter allocations, demonstrating that placement matters beyond total strength. The approach is model-agnostic, low-cost, and complementary to parameter-efficient fine-tuning (LoRRA), providing a practical knob for eliciting truthful reporting from existing capabilities.

5 Limitations

Our study has several limitations. First, the method requires activation-level access to language model’s layers and the ability to inject per-layer steering strengths, which restricts its applicability to open-weight models. Second, our evaluation relies on an external LLM judge with specific prompting strategies, which introduces potential sensitivity to both judge selection and prompt design. While this approach enables scalable evaluation, incorporating multiple judges or human evaluation would provide more robust validation of the reported improvements. Third, our experiments focus primarily on the MASK benchmark, which limits the generalizability of our findings. Future work should validate the approach across a broader range of safety benchmarks, such as Machiavelli [Pan et al., 2023].

Acknowledgement

We would like to thank MARS (Mentorship for Alignment Research Students) for mentorship and financial support. We would also like to thank the Center for AI Safety for providing computational resources.